

Maelstrom Cheat Sheet

M1: Clojure Runtime & Topology

Card 1: Clojure JVM Runtime

Runs on JVM, managing spawned node processes and intercepting network links for testing consistency.

Card 2: STDIN/STDOUT Protocol

Uses newline-delimited JSON. Nodes must never write debug outputs to stdout, only to stderr.

Card 3: Node-Node Communication

Redirects inter-node messages in-memory, bypassing physical socket I/O for speed.

Card 4: Loop & State Concurrency

Clojure Agents and Atoms handle state asynchronously. Node count is recommended below 20.

Card 5: JSON Schema RPC

Ensures strict RPC semantics using 'msg_id' and 'in_reply_to' matching over stdin/stdout.

M2: Jepsen Engine & Harness

Card 6: Knossos Linearizability

Verifies single-key histories via WGL path analysis. Use '-time-limit' to prevent OOM errors.

Card 7: Elle Transaction Checker

Constructs transaction graphs to detect G1/G2 anomalies in multi-key isolation models.

Card 8: Generator Pipeline

Defines client workloads and concurrency via Clojure thread configurations.

Card 9: Checker Engine

Parses history log files to generate 'results.edn' validation metrics.

Card 10: Client Proxy Concurrency

Maps client proxies to physical OS threads for latency assessment under chaos.

M3: Consistency & Theorems

Card 11: Causal Consistency

Causally related writes must be seen in the same order. Uses vector clocks in headers.

Card 12: Read Uncommitted / Committed

Differentiates dirty reads from MVCC visibility. Validated by searching for G1a/G1b cycles.

Card 13: Repeatable Read

Ensures reads yield identical results throughout a transaction using lock policies.

Card 14: Serializability

Demands execution history be equivalent to some serial ordering. Checks for G2 graph cycles.

Card 15: Monotonic Read/Write

Prevents reading stale versions relative to previous client reads in a session.

M4: Distributed Storage

Card 16: Lin-KV Strong Consistency

A built-in linearizable key-value store endpoint for sharing test states.

Card 17: Seq-KV Sequential Consistency

Provides sequential ordering, offering higher throughput than 'lin-kv'.

Card 18: Lww-Register Conflict Resolution

Merges register state using highest timestamp. Skewed physical clocks can lead to data loss.

Card 19: PN-Counter State Merge

CRDT counter combining positive/negative vectors using element-wise maximum check.

Card 20: Replicated Log

Requires majority quorum ('N/2+1') approval before commits to prevent split-brain.

M5: Network Chaos & Faults

Card 21: Network Partition

Partitions cluster into unreachable groups to test split-brain behaviors.

Card 22: Packet Drop Simulation

Drops packets at a custom '-drop-rate' to verify network retry safety.

Card 23: Latency Inflation & Reordering

Simulates congestion delays and reordering using '-latency' and '-delta' bounds.

Card 24: Nemesis Coordinator

Manages fault injection state machines and runs chaos schedules.

Card 25: Recovery & Reconnection

Heals routing paths and tests catch-up synchronizations from healed nodes.

M6: Diagnostics & CLI

Card 26: Gnuplot Latency & Throughput Graphs

Visualizes raw performance over time using 'latency-raw.png' scatterplots.

Card 27: History Log Analysis

Inspects client event history records under 'store/latest/history.txt' directly.

Card 28: CLI Parameter Reference

Usage: './maelstrom test -w txn-list -bin ./my_node -node-count 5 -rate 100'